Yachay Tech University

School of Mathematical and Computational Sciences

Stochastic Processes: Midterm Challenge

Evidence-Based Band-Pass Filter Design for Real Audio

Total: 10 points **Teams of exactly two students**

The challenge

Recover a justified frequency band from one real recording. Estimate suitable cutoff frequencies from evidence, design low-order continuous- and discrete-time filters, and demonstrate what is preserved, rejected, and distorted. Compare two digital implementations under the original recording conditions and controlled additive and multiplicative noise. **Required outputs:** a reproducible Python notebook (Colab-compatible), a concise PDF report, and a five-minute pitch presentation. A live oral pitch and questions will take place if class time allows. Submission date and working time will be announced by the instructor.

Choose one signal

Use one of the following four recordings throughout the challenge. Links point to downloadable OGG files; the `librosa` keys give a second access route (page 6). Use the full clip or justify a contiguous excerpt with enough target activity and background context.

A. Speech in a real recording — `libri1`**.** Preserve intelligibility while rejecting unnecessary low- and high-frequency content. Discuss the conflict between preserving vowels and consonants; the largest spectral peak alone does not define the useful band.
Download: narration recording (OGG).

B. Robin call — `robin`**.** Preserve the bird's time-varying call and reject surrounding spectral content. Separate target activity from background using a spectrogram; a single average spectrum may conceal the call's frequency excursions.
Download: robin recording (OGG).

C. Humpback whale — `humpback`**.** Preserve a selected vocalization band amid underwater background sound. Justify the selected events and whether one fixed band can retain their changing tonal structure.
Download: humpback whale recording (OGG).

D. Trumpet — `trumpet`**.** Preserve the main musical line and a justified set of harmonics. Explain the trade-off between rejecting unwanted content and altering timbre; not every high-frequency harmonic is noise.
Download: trumpet recording (OGG).

General conventions

Use $x(t), x[n]$ for inputs, $y(t), y[n]$ for outputs, and $u(t), u[n]$ for unit steps. Assume zero initial conditions. Use Ω in rad/s, ω in rad/sample, and f in Hz; $\omega = 2\pi f/f_s$. Clearly distinguish the original recording, synthetic corruption, and filter output. Derivations must accompany software results.

1. Estimate the band and design two competing filters

1.1 Signal investigation and specifications

Report the source, sample rate, duration, channel handling, excerpt timestamps, and any resampling. Preserve the native sample rate unless a change is justified; apply anti-alias filtering before downsampling. Show the waveform, Welch power spectral density (PSD), and STFT spectrogram with units and stated parameters.

Identify target-active and background-dominated intervals where available. Propose $0 < f_L < f_H < f_s/2$, using listening, time-frequency evidence, and the target's physical meaning. Compare at least two candidate bands before selecting one. Do not infer the whole band solely from the highest FFT peak or copy standard speech/music cutoffs without evidence.

Give a specification table: desired -3 dB band edges, an interior passband, one stopband below and one above it, allowed passband variation, desired stopband attenuation, and maximum acceptable delay. Choose feasible specifications for the order limits below. Report unmet specifications honestly and discuss the compromise; a justified low-order limitation is not a failed experiment.

1.2 Mandatory route A: continuous Butterworth to digital IIR

Start with a normalized Butterworth low-pass prototype of order $N = 1$ or $N = 2$. Transform it to a band-pass filter, giving final analog order $2N = 2$ or 4. Derive the transformation and show numerical coefficients or second-order factors.

$$\Omega_L = 2f_s \tan\left(\frac{\pi f_L}{f_s}\right), \qquad \Omega_H = 2f_s \tan\left(\frac{\pi f_H}{f_s}\right), \qquad B = \Omega_H - \Omega_L, \quad \Omega_0 = \sqrt{\Omega_L \Omega_H},$$

$$H_c(s) = H_{\mathrm{LP}}\left(\frac{s^2 + \Omega_0^2}{Bs}\right), \qquad H_A(z) = H_c\left(2f_s \frac{1 - z^{-1}}{1 + z^{-1}}\right).$$

Explain why both desired digital edges are prewarped. Analog edges are Ω_L, Ω_H in rad/s, not generally $2\pi f_L, 2\pi f_H$. Include a second conversion without prewarping as a diagnostic and quantify its digital edge errors. SciPy's `bilinear` does not perform prewarping for you [2].

1.3 Mandatory route B: direct digital alternative

Choose **one** comparator with the same target band and sampling rate:

- **FIR:** a window-designed linear-phase band-pass with an odd number of taps, at most 65 (order at most 64). State the window, tap count, delay, and transition-width limitations.
- **IIR:** a directly designed band-pass, final order at most 4. Use a different family or order from route A (for example, Chebyshev I) and explain the choice. An identical digital Butterworth design can verify route A, but does not constitute a distinct comparator.

Use stable second-order sections for IIR filtering. FIR filters need more coefficients than IIR filters; keep their order bounded and explain the unequal-order trade-off. A cascade of independently designed high- and low-pass filters need not equal the Butterworth band-pass and must not be presented as identical.

2. Mathematical models, stability, and verification

2.1 Transfer functions and governing equations

For route A, present the numerical analog transfer function and its differential equation:

$$H_c(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{P} a_k s^k} \quad \Longrightarrow \quad \sum_{k=0}^{P} a_k \frac{d^k y}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x}{dt^k}.$$

For both digital designs, normalize $a_0 = 1$ and present

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{P} a_k z^{-k}}, \qquad y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{P} a_k y[n-k].$$

Show all IIR coefficients; for an FIR, give the coefficient array in the notebook and its finite-sum model in the report. A direct FIR comparator does not require an invented continuous-time differential equation. Implement one digital recursion explicitly and compare it against a standard routine on a short test sequence; report maximum absolute error.

2.2 Stability and causality

Plot analog poles and zeros in the s plane and digital poles and zeros in the z plane with the unit circle. Verify $\Re(p_c) < 0$ and $|p_d| < 1$ for causal IIR systems and state their causal regions of convergence. Explain the mapping

$$z = \frac{1 + s/(2f_s)}{1 - s/(2f_s)}$$

and why it preserves left-half-plane stability. Report the largest digital pole modulus and the analog pole closest to the imaginary axis. For the FIR, establish BIBO stability through its finite, absolutely summable impulse response. Pole locations alone must be accompanied by numerical values and reasoning.

2.3 Frequency- and time-domain evidence

(a) Plot analog Bode magnitude and unwrapped phase, and the digital magnitude, phase, and group delay for A and B. Mark target and measured edges, interior passband, and stopbands. Report gain variation, stopband attenuation, and delay with units. For window-designed FIR filters, a `firwin` cutoff is nominally a -6 dB point [4]; measure actual -3 dB crossings instead of equating these conventions.

(b) Compare analog and digital responses on both a common physical-Hz axis and the correctly warped axis $\Omega = 2f_s \tan(\omega/2)$. Explain why equal physical frequencies need not yield identical responses. Exclude Nyquist from the warped plot.

(c) Plot impulse and step responses for H_c, H_A, and H_B. Use a true continuous impulse-response routine, not a unit-height sampled pulse. Explain the expected steady-state response to a step for a band-pass filter.

(d) Test a three-tone input with one tone below, one inside, and one above the selected band. Compare predicted and measured steady-state gains. Simulate the analog system with an explicitly stated input interpolation; halve the integration/output spacing to check convergence. Explain discrepancies with digital outputs caused by warping and sampling.

Use causal one-pass filtering for assessed comparisons. Optional `filtfilt` results must be separate: forward–backward filtering changes the effective magnitude response and is noncausal.

3. Noise robustness and meaningful metrics

3.1 Controlled corruption in addition to inherent noise

Let $r[n]$ be the selected uncorrupted-by-you recording, including its inherent background. Construct

$$x_{\alpha,\beta}[n] = r[n]\big(1 + \beta m[n]\big) + \alpha v[n],$$

where $v[n]$ is zero-mean, unit-RMS additive noise and $m[n]$ is independent zero-mean, unit-RMS *slowly varying* noise. Generate m by smoothing/filtering random noise; state its bandwidth and motivate it as random gain fluctuation. Explain that multiplication creates sidebands and signal-dependent noise that an LTI band-pass cannot necessarily remove.

Use white Gaussian v as the common benchmark. Additionally test one context-appropriate colored additive noise (for example, low-frequency rumble or broadband hiss) and explain its PSD. For each generated realization, demean and normalize the noise before scaling. Use five independent, recorded random seeds and the same noisy input for both filters.

Case	Additive level	Multiplicative level
Baseline	None	$\beta = 0$
Additive only	Input SNR 20 and 5 dB	$\beta = 0$
Multiplicative only	None	$\beta = 0.10$ and 0.30
Combined	Input SNR 5 dB	$\beta = 0.30$

Use the colored-noise variant at least for the 5 dB additive-only case. Set $\alpha = \mathrm{RMS}(r)10^{-\mathrm{SNR}_{\mathrm{add}}/20}$ after unit-RMS normalization of v. Report the achieved combined corruption SNR; its value is not generally 5 dB. Retain floating-point signals without clipping and do not renormalize individual test cases before metrics.

3.2 Report separate measures of noise and distortion

No clean source is supplied. Do not claim that a PSD ratio or comparison to r measures the unknown true SNR of the field recording. For synthetic corruption $e = x - r$, report

$$\mathrm{SNR}_{\mathrm{in}}^{\mathrm{ref}} = 10\log_{10}\frac{\sum r[n]^2}{\sum e[n]^2}, \qquad \mathrm{SNR}_{\mathrm{out}}^{\mathrm{ref}} = 10\log_{10}\frac{\sum (Hr)[n]^2}{\sum (He)[n]^2}.$$

Here H is the same causal filter applied separately to r and e, with identical zero initial conditions and evaluation interval. Report ΔSNR; this quantifies rejection of *added* corruption relative to the filtered recording and does not prove preservation of the target.

Also report delay-aligned RMSE and correlation between Hx and r, explicitly labelled *recording-reference fidelity*, alongside in-band energy retention on Hr and attenuation of e in each stopband. Define PSD integration bands and ratios. Filtering may intentionally remove content in r, so discuss this metric's limitations. Estimate one alignment lag per filter from the uncorrupted recording and reuse it for all noisy trials; retain unaligned results or state the compensation clearly.

Report baseline results and mean $\pm$ standard deviation over seeds for noisy cases. State transient exclusion, alignment, effective sample count, and any zero-energy/undefined metrics. Listening and spectrogram evidence must accompany the metric table. Do not promise positive SNR improvement when signal and noise overlap.

4. Reproducible demonstration and submission

4.1 Notebook interface (required); API (optional)

The notebook must run from a fresh Colab session, download the chosen recording, and regenerate every figure and table. Provide a simple selector for filter A/B and noise condition, with four labelled audio players:

- Original or currently selected corrupted input (clearly labelled).
- **High-pass only:** pass frequencies above f_L; low frequencies are suppressed.
- **Low-pass only:** pass frequencies below f_H; high frequencies are suppressed.
- **Band-pass:** retain the selected middle band using the assessed A or B design.

Always retain a player for the untouched recording. State the high-/low-pass designs and their orders; these players are diagnostic comparisons, not substitutes for the two band-pass designs. Use one shared playback scaling factor across compared outputs and disable automatic per-player normalization. Export WAV files without clipping. An optional API may expose the same processing options and return audio plus metrics, but does not replace the notebook and requires no public deployment.

4.2 Main deliverables

1. **Python notebook (.ipynb):** installation/import cell, versions, data retrieval and attribution, parameters and seeds, derivations or linked explanations, all experiments, audio interface, and export cells. Include a Colab link if used; the submitted file must remain independently runnable.
2. **PDF report:** approximately 6–8 pages excluding references and coefficient appendices. Include the band-selection argument, models, stability, comparison figures, metric tables, limitations, and final recommendation. Include LaTeX source if used. State each student's contributions.
3. **Five-minute pitch:** submit 4–6 slides as PDF. Suggested timing: target and evidence (45 s), designs and stability (75 s), listening comparison (60 s), metrics and trade-offs (75 s), recommendation (45 s). Both students contribute. Deliver the pitch live, with brief questions, if time allows; otherwise submit a five-minute narrated recording. Grading weight remains unchanged.

Assessment rubric — Total: 10 points

Criterion	Evidence for full credit	Points
Signal and band selection	Verified source, PSD/STFT and listening evidence, candidate bands, justified specifications.	1.50
Models and discretization	Correct analog Butterworth development, prewarping/Tustin, equations and coefficients, distinct digital comparator.	2.00
Stability and verification	Pole/zero reasoning, Bode/phase/delay, measured edges, impulse/step and three-tone checks.	1.50
Noise and metrics	Appropriate additive/multiplicative models, matched seeded trials, valid SNR labels, distortion and uncertainty analysis.	2.00
Notebook and demonstration	Fresh-session reproducibility, four listening modes, fair scaling, parameter control and exports.	1.25
Report	Clear evidence, cited sources, critical conclusions and contributions.	0.75
Five-minute pitch	Clear technical argument, concise demo, shared participation and defensible recommendation.	1.00
Total		**10.00**

Partial credit per criterion: 100% complete, correct, justified; 75% sound with minor gaps; 50% substantial progress with major gaps; 25% limited relevant progress; 0% missing or unsupported. A carried arithmetic error does not erase credit for a correct subsequent method. Honest identification of infeasible low-order specifications earns credit for analysis.

Data access and references

Minimal acquisition example (not a filter solution)

The following cell retrieves any one of the four recordings without a dataset account. Use `sr=None` to preserve its native rate. Record licensing and attribution using the metadata function.

```
# Run once in a fresh Colab session:
%pip -q install librosa soundfile scipy matplotlib ipywidgets
import librosa
from IPython.display import Audio, display

key = "robin" # choose: libri1, robin, humpback, trumpet
path = librosa.ex(key)
r, fs = librosa.load(path, sr=None, mono=True)
print("sample␣rate:", fs, "samples:", len(r),
      "duration␣(s):", len(r) / fs)
librosa.util.example_info(key)
display(Audio(r, rate=fs, normalize=False))
```

For a fair listening comparison, compute a common scale from the largest absolute sample across all outputs in that comparison, apply it to every player, and retain the unscaled arrays for calculations. If the automatic download fails, use the direct OGG link on page 1 and load the local file. Preserve attribution and comply with the recording's stated license.

Availability check

All four direct links were downloaded and decoded during preparation on 16 October 2026. No login was required. Availability at a future date cannot be guaranteed; keep a local copy of the selected recording for reproducibility. The checked native-file properties are listed below.

Key	Sample rate (Hz)	Duration (s)	Channels
libri1	22,050	14.840	1
robin	22,050	2.699	1
humpback	22,050	64.809	1
trumpet	22,050	5.333	1

References and implementation guidance

[1] librosa, *Example files*: recording catalogue, access keys, and per-recording metadata.
https://librosa.org/doc/main/recordings.html

[2] SciPy, `signal.bilinear`: Tustin substitution and the absence of automatic prewarping.
https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.bilinear.html

[3] SciPy, `signal.butter`: analog/digital designs, critical-frequency convention, order doubling for band-pass filters, and SOS guidance.
https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.butter.html

[4] SciPy, `signal.firwin`: window-based linear-phase FIR design and half-amplitude cutoff convention.
https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.firwin.html

[5] SciPy, *Signal processing*: frequency responses, LTI simulation, Welch PSD, filtering, and related functions.
https://docs.scipy.org/doc/scipy/reference/signal.html

Final decision: recommend one design for your selected target, supported by measured selectivity, stability, distortion, latency, computational cost, noise robustness, and listening evidence. There is no single prescribed cutoff pair and no requirement that one filter win every comparison.